Chapter 3 - Rings and Moduls of Fractions

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Exercise 1

(i) ($\Rightarrow$): Let M be generated by $\{m_1, m_2, \ldots, m_n\}$, for $\forall i \in [1, n] \cap \mathbb{Z}$, we have

$$\forall r, t \in S, \exists s \in S, \text{s.t. } m_i s(r-t) = 0, \text{i.e. } m_i s = 0$$

Denote the s above as s_i. Since M is finitely-generated, we can let

$$s = s_1 s_2 \ldots s_n$$

And have $sM = 0$.

(ii) ($\Leftarrow$): When such s exist, then for $\forall \frac{m_1}{s_1}, \frac{m_2}{s_2} \in S^{-1}M$,

$$s(s_1 m_2 - s_1 m_2) = 0 \implies \frac{m_1}{s_1} = \frac{m_2}{s_1}$$

So $S^{-1}M = 0$.

$\square$

Exercise 2

By *Proposition 1.9*, the problem is equivalent to prove that for any $\frac{a}{s} \in S^{-1}A$, we have

$$\forall \frac{b}{t} \in S^{-1}A, 1 - \frac{a}{s}\frac{b}{t} \text{is a unit in } S^{-1}A$$

Since $1 + \mathfrak{a}$ is a multiplicative set, $\exists c \in \mathfrak{a}, \text{s.t. } st = 1 + c$. Now

$$\begin{aligned} 1 - \frac{a}{s}\frac{b}{t} &= 1 - \frac{ab}{st} \\ &= 1 - \frac{ab}{1+c} \\ &= \frac{1+c-ab}{1+c} \end{aligned}$$

And we can easily see $\frac{1+c}{1+c-ab} = 1 + \frac{ab}{1+c-ab} \in S^{-1}A$. So the inverse of $1 - \frac{a}{s}\frac{b}{t}$ is $1 + \frac{ab}{1+c-ab}$. $\square$

Exercise 3

Define

$$f : (ST)^{-1}A \longrightarrow U^{-1}(S^{-1}A)$$

by mapping

$$\frac{a}{st} \longmapsto \frac{\frac{a}{s}}{\frac{t}{1}}$$

We'll prove f is an isomorphism:

(i) Surjection: Trivial.

(ii) Injection:

$$\frac{\frac{a_1}{s_1}}{\frac{t_1}{1}} = \frac{\frac{a_2}{s_2}}{\frac{t_2}{1}}$$

$\Leftrightarrow$

$$\exists t \in T, \text{s.t. } \frac{t}{1}\left(\frac{a_1 t_2}{s_1} - \frac{a_2 t_1}{s_2}\right) = 0$$

$\Leftrightarrow$

$$\exists t \in T, \text{s.t. } t(a_1 t_2 s_2 - a_2 t_1 s_1) = 0$$

$\Leftrightarrow$

$$\frac{a_1}{s_1 t_1} = \frac{a_2}{s_2 t_2}$$

So f is injective.

(iii) Homomorphism: Trivial.

□

Exercise 4

Define

$$f : S^{-1}B \longrightarrow T^{-1}B$$

by

$$\frac{b}{s} \longmapsto \frac{b}{f(s)}$$

We'll prove it's an isomorphism:

(i) Homomorphism: Trivial

(ii) Surjection: Trivial

(iii) Injection:

$$\frac{b_1}{f(s_1)} = \frac{b_2}{f(s_2)}$$

$\Leftrightarrow$

$$\exists s \in S, \text{s.t. } f(s)\left(b_1 f(s_2) - b_2 f(s_1)\right) = 0$$

$\Leftrightarrow$

$$\frac{b_1}{s_1} = \frac{b_2}{s_2}$$

□

Exercise 5

(i) By *Proposition 3.11-(v)*, the operation S^{-1} commutes with the formations of radicals. So $\forall \mathfrak{p} \subseteq A$,

$$(\text{Nil}(A))_{\mathfrak{p}} = \text{Nil}(A_{\mathfrak{p}}) = 0$$

Therefore by *Proposition 3.8*, $\text{Nil}(A_{\mathfrak{p}}) = 0$.

(ii) No. Counterexample given here

Exercise 6

(i) Σ is nonempty, therefore has maximal elements by *Zorn's Lemma*.

(ii) (I) When $S \in \Sigma$ is maximal, we'll prove $A\backslash S$ is a minimal prime ideal. $\forall x, y \in A\backslash S_0$, let

$$S_x = \left\{sx^n \mid s \in S, n \in \mathbb{Z}^+ \cup \{0\}\right\}, S_y = \left\{sy^n \mid s \in S, n \in \mathbb{Z}^+ \cup \{0\}\right\}$$

Since $S \in \Sigma$ is maximal and $S_x, S_y \supsetneq S$, we have $S_x, S_y \notin \Sigma, 0 \in S_x, 0 \in S_y$ Therefore $\exists s_1, s_2 \in S, n_1, n_2 \in \mathbb{Z}^+$, such that

$$s_1 x^{n_1} = s_2 x^{n_2} = 0$$

And we can see

$$s_1 s_2 (x+y)^{n_1+n_2-1} = 0, s_1 s_2 (xy)^{n_1 n_2} = 0$$

Which is to say $x+y, xy \in A\backslash S$, and we can easily obtian $A\backslash S$ is an ideal. The prime property is trivial, and the maximal property can be obtained directly from *(II)* and the maximal property of S.

(II) When $A\backslash S$ is a minimal prime ideal of A: Denote $I = A\backslash S$. For $\forall x, y \in S$, we have $x, y \notin I$. If $xy \in I$, by the prime property of I we will immediately obtian $x \in I$ or $y \in I$, which is a contradictory. So $xy \notin I, xy \in S$, S is a multiplicatively closed subset.

Also, if S is not maximal in Σ, the one in Σ strictly contains S will correspond to a smaller prime ideal in $A\backslash S$, which contradicts with the minimal property of I.

$\square$

Exercise 7

(i) (I) $(\Leftarrow)$: When $A\backslash S$ is a union of prime ideals, $x \in A\backslash S$ or $y \in A\backslash S$ will immediately implies $xy \notin S$. So $xy \in S$ will imply $x, y \in S$.

(II) $(\Rightarrow)$: We'll prove that any elements in $A\backslash S$ was contained in a prime ideal which is disjoint to S: $\forall a \in A\backslash S, (a) \cap S \subseteq \{0\}$, otherwise if $a^n = a \cdot a^{n-1} \in S$, we'll have $a \in S$, contradictory.

Now we consider Σ be the set of all ideals containing (a) and being disjoint to S. By *Zorn's Lemma*, we can pick out a maximal element, namely $\mathfrak{p} \subseteq \Sigma$. We'll prove $\mathfrak{p}$ is prime:

Suppose we have $xy \in \mathfrak{p}, x, y \notin \mathfrak{p}$. Since $\mathfrak{p}$ is maximal and $\mathfrak{p} + (x), \mathfrak{p} + (y) \supsetneq \mathfrak{p}$, we know both of $\mathfrak{p} + (x), \mathfrak{p} + (y)$ intersects with S. Therefore $\exists s_1, s_1 \in S$ such that $\exists p_1, p_2 \in \mathfrak{p}, a_1, a_2 \in A$,

$$s_1 = p_1 + a_1 x, s_2 = p_2 + a_2 x$$

Then we have

$$(s_1 - p_1) s_2 \in S$$

whereas

$$\begin{aligned}(s_1 - p_1) s_2 &= a_1 x (p_2 + a_2 y) \\ &= a_1 p_2 x + a_1 a_2 xy \in \mathfrak{p}\end{aligned}$$

Which means $(s_1 - p_1) s_2 \in S \cap \mathfrak{p}$, contradicting with the construction of $\mathfrak{p}$.

$\square$

(ii) By *(i)*, $\overline{S}$ defined by the complement in A of union of prime ideals disjoint from S is indeed a saturated multiplicative closed subset of A.

We only have to prove $\overline{S}$ is the unique minimal one, which is trivial since the family of sets, which consists of prime ideals disjoint from S, has a unique maximal element (final object, where morphisms taken under inclusion).

(iii)

$$\mathfrak{p} \cap S \subsetneq \{0\}$$

$\Leftrightarrow$

$$\exists p \in \mathfrak{p}, a \in \mathfrak{a}, \text{s.t. } 1 + p = a$$

$\Leftrightarrow$

$$\mathfrak{a} + \mathfrak{p} = (1)$$

$\Leftrightarrow$

$$\mathfrak{a}, \mathfrak{p} \text{ are coprime}$$

Conversely, it's easy to prove that $\mathfrak{a}, \mathfrak{p}$ being coprime will implies $\mathfrak{p} \cap S \subsetneq \{0\}$. Therefore $\overline{S}$ is the complement of the union of all prime ideals which are not coprime with $\mathfrak{a}$.

Exercise 8
(UNFINISHED)

(I) *(i)* $\Rightarrow$ *(ii)*: $\forall t \in T, \phi(\frac{t}{1}) = \frac{t}{1}$ is a unit in $T^{-1}A$. Since ϕ is bijective, the preimage of the inverse of $\frac{t}{1} \in T^{-1}A$ is also the inverse of $\frac{t}{1} \in S^{-1}A$.

(II) *(ii)* $\Rightarrow$ *(iii)*: $\forall t \in T$, $\frac{t}{1}$ is a unit in $S^{-1}A$ and we denote $\frac{a}{s} = \left(\frac{t}{1}\right)^{-1}$. Therefore $\exists s_0 \in S$, s.t. $s_0(at - s) = 0$, which indicate

$$(s_0 a)\, t = s_0 s \in S$$

Thus we can let $x = s_0 a$.

(III) *(iii)* $\Rightarrow$ *(iv)*: Suppose $\exists t \in T$, s.t. t is contained in a prime ideal $\mathfrak{p}$ which is disjoint from S. By *(iii)*, $\exists x \in A$, s.t. $xt \in S$. However, $t \in \mathfrak{p} \Rightarrow xt \in \mathfrak{p} \Rightarrow xt \in S \cap \mathfrak{p}$, contradictory.

(IV) *(iv)* $\Rightarrow$ *(v)*: Otherwise T cannot be contianed in $\overline{S}$.

(V) *(v)* $\Rightarrow$ *(i)*:

Exercise 9

(I) We follow from the hint: Denote the family of all multiplicatively closed set which doesn't contains 0 by Σ, the problem is equivalent to prove that all the maximal elements in Σ must contains S.

We prove by contradictory and suppose there exist a maximal element $S \in \Sigma$ such that $\exists a \in S_0 \backslash S$. Let $S_a = \{sa^n \mid s \in S, n \in \mathbb{Z}^+ \cup \{0\}\}$, which is a multiplicatively closed set strictly contains S. Because S is maximal in Σ, $S_a \notin \Sigma, 0 \in S_a$. Hence, $\exists n \in \mathbb{Z}^+$, s.t.

$$sa^n = 0$$

However, from $a \in S_0 \backslash S$ we know a^n is a non-zero-divisor, i.e. $s = 0$, which is absurd. Hence the proof was done. $\square$

(II)
   (i) Homomorphism $A \longrightarrow S_0^{-1}A$ is injective iff $\forall s \in S_0, a \in A \backslash \{0\}, sa \neq 0$. This is to say S_0 consists of non-zero-divisors.
   (ii) Every element in $A \backslash S_0$ will be mapped to a zero-divisor while every element in S_0 will be mapped to a unit, obviously.
   (iii) Injectivity is obtained from *(i)*, and the preimage of $\frac{a}{s} \in S^{-1}A$ is $as^{-1} \in A$, obviously.

$\square$

Remark 1. *From* ***Exercise 3.6, 3.9****, we can see saturated sets and sets made up of prime-ideals-union acts as complements to each other. And if we let S be maximal, the corresponding $A \backslash S$ will be left to be only one prime idea.*

Exercise 10

(i) We need to prove that for any $S^{-1}A$-module $S^{-1}M$ with short exact sequence of $S^{-1}A$-module

$$N' \xrightarrow{f} N \xrightarrow{g} N''$$

we have

$$N' \otimes_{S^{-1}A} S^{-1}M \xrightarrow{f \otimes 1} N \otimes_{S^{-1}A} S^{-1}M \xrightarrow{g \otimes 1} N'' \otimes_{S^{-1}A} S^{-1}M$$

First, since the restriction below only affect the inner multiplicative strcture while preserving the image and kernel, we have

$$N' \mid_A \xrightarrow{f|_A} N \mid_A \xrightarrow{g|_A} N'' \mid_A$$

being a short exact sequence of A-module. Also, since A is absolutely flat,

$$N' \mid_A \otimes_A M \xrightarrow{f_A \otimes 1_A} N \mid_A \otimes_A M \xrightarrow{g_A \otimes 1_A} N'' \mid_A \otimes_A M$$

is exact.

Hence by *Proposition 3.3*, we only need to prove that for any $S^{-1}A$-module N we have

$$S^{-1}\left(N \mid_A \otimes_A M\right) \cong N \otimes_{S^{-1}A} S^{-1}M$$

First, for any $S^{-1}A$-module N, $S^{-1}\left(N \mid_A\right) \cong N$ (we can prove by defining the cannonical mapping and state is's surjective and injective). So

$$\begin{aligned} S^{-1}\left(N \mid_A \otimes_A M\right) &\cong S^{-1}(N \mid_A) \otimes_{S^{-1}A} S^{-1}M \\ &\cong N \otimes_{S^{-1}A} S^{-1}M \end{aligned}$$

(ii)